AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 9 · WORKBOOK

Integration tests and fixtures

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete integration-fixture and interaction record.

Keep requests, fixture modes, expected/actual results, call traces and limits together.

Keep one habit: Recall → W1–W3/W5 → independent missing-evidence case → project interaction tests → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Integration test**s exercise connected components and failure propagation. **Fixture**s supply controlled **input** records and mocked service outcomes.

- Trace request validation, session permission, lookup and **output** validation together. Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup.

Testing model-dependent behaviour

> **My remembered idea, example and correction**

Least privilege and tool scopes

> **My remembered idea, example and correction**

Typed contracts and schemas

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main integration fixture: S student-visible source S/roomB/capacity 12; T teacher-only source T/STAFF/capacity 4. Trusted caller student. Correct request validation → authorization → lookup → local draft stub → response validation. Student request T returns denied with no lookup calls. Deliberately broken ordering calls T before denying, so final response alone misses the hidden boundary failure.

  Allowed S normal fixture answers S/B/12 and records lookup S. Missing or failing storage returns review before drafting. Preserve both response and event/call trace. Separate catalogue fixture M=map, N=notebook: requested N must yield source N/notebook; adapter M/map is a wrong-ID connection and should not be published as the requested answer.

  Mocks test declared interactions and failure propagation; no live network/provider, model generation or complete production guarantee is supplied.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main integration fixture: S student-visible source S/roomB/capacity 12; T teacher-only source T/STAFF/capacity 4. Trusted caller student. Correct request validation → authorization → lookup → local draft stub → response validation. Student request T returns denied with no lookup calls. Deliberately broken ordering calls T before denying, so final response alone misses the hidden boundary failure.

  Allowed S normal fixture answers S/B/12 and records lookup S. Missing or failing storage returns review before drafting. Preserve both response and event/call trace. Separate catalogue fixture M=map, N=notebook: requested N must yield source N/notebook; adapter M/map is a wrong-ID connection and should not be published as the requested answer.

  Mocks test declared interactions and failure propagation; no live network/provider, model generation or complete production guarantee is supplied.

- A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main integration fixture: S student-visible source S/roomB/capacity 12; T teacher-only source T/STAFF/capacity 4. Trusted caller student. Correct request validation → authorization → lookup → local draft stub → response validation. Student request T returns denied with no lookup calls. Deliberately broken ordering calls T before denying, so final response alone misses the hidden boundary failure.

  Allowed S normal fixture answers S/B/12 and records lookup S. Missing or failing storage returns review before drafting. Preserve both response and event/call trace. Separate catalogue fixture M=map, N=notebook: requested N must yield source N/notebook; adapter M/map is a wrong-ID connection and should not be published as the requested answer.

  Mocks test declared interactions and failure propagation; no live network/provider, model generation or complete production guarantee is supplied.

- A report omits this qualification: A mock verifies declared interactions, not real provider availability or all runtime limits. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main integration fixture: S student-visible source S/roomB/capacity 12; T teacher-only source T/STAFF/capacity 4. Trusted caller student. Correct request validation → authorization → lookup → local draft stub → response validation. Student request T returns denied with no lookup calls. Deliberately broken ordering calls T before denying, so final response alone misses the hidden boundary failure.

  Allowed S normal fixture answers S/B/12 and records lookup S. Missing or failing storage returns review before drafting. Preserve both response and event/call trace. Separate catalogue fixture M=map, N=notebook: requested N must yield source N/notebook; adapter M/map is a wrong-ID connection and should not be published as the requested answer.

  Mocks test declared interactions and failure propagation; no live network/provider, model generation or complete production guarantee is supplied.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **integration test**s and **fixture**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **integration test**s and **fixture**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Integration test**s and **fixture**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Integration test**s and **fixture**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

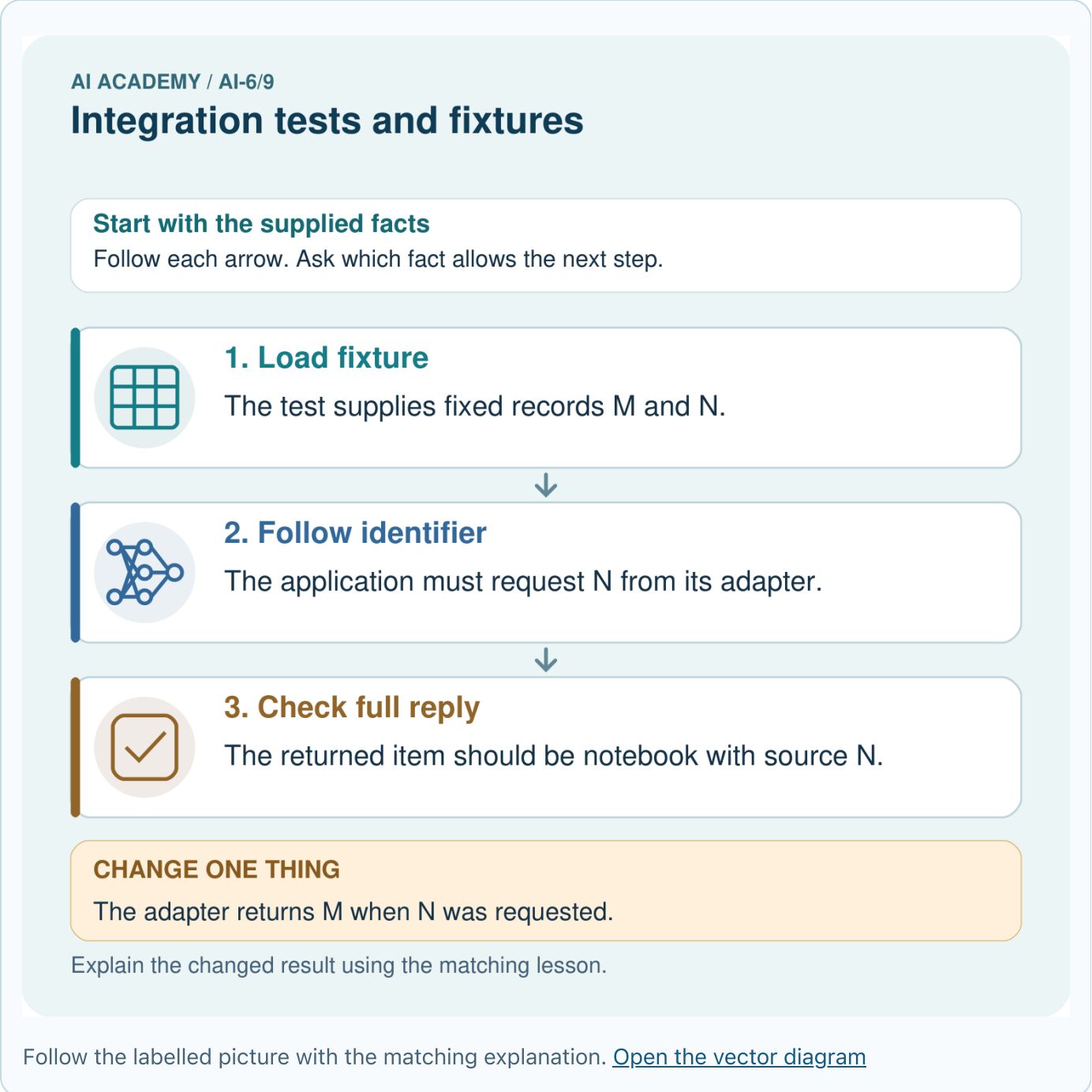
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Catch a wrong adapter connection. A service reads a fake catalogue **fixture** and answers a request for item N.
- **Fixture** M="map", N="notebook". No external network is involved.
- Follow request N through the **fixture** and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Catch a wrong adapter connection. A service reads a fake catalogue **fixture** and answers a request for item N.
- Which labelled step matches this detail? The returned item should be notebook with source N.
- 1. Load **fixture**
- 2. Follow identifier

- 3. Check full reply

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check full reply The returned item should be notebook with source N. **Integration test**ing checks how components connect.

Each component can work alone while their identifier exchange is wrong.

Changed-case question and optional hint

- Changed-case practice: The adapter returns M when N was requested. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. Fail the integration check: map is the wrong item.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Fail the integration check: map is the wrong item. The request now reaches **evidence** for another identifier.

Repair and retain the record

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Review Typed contracts and schemas (ai-6/3). Goal: Explain the **mechanism** and terminology of typed contracts and schemas. Short practice: Compare A's count 4 with B's count "4".

Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

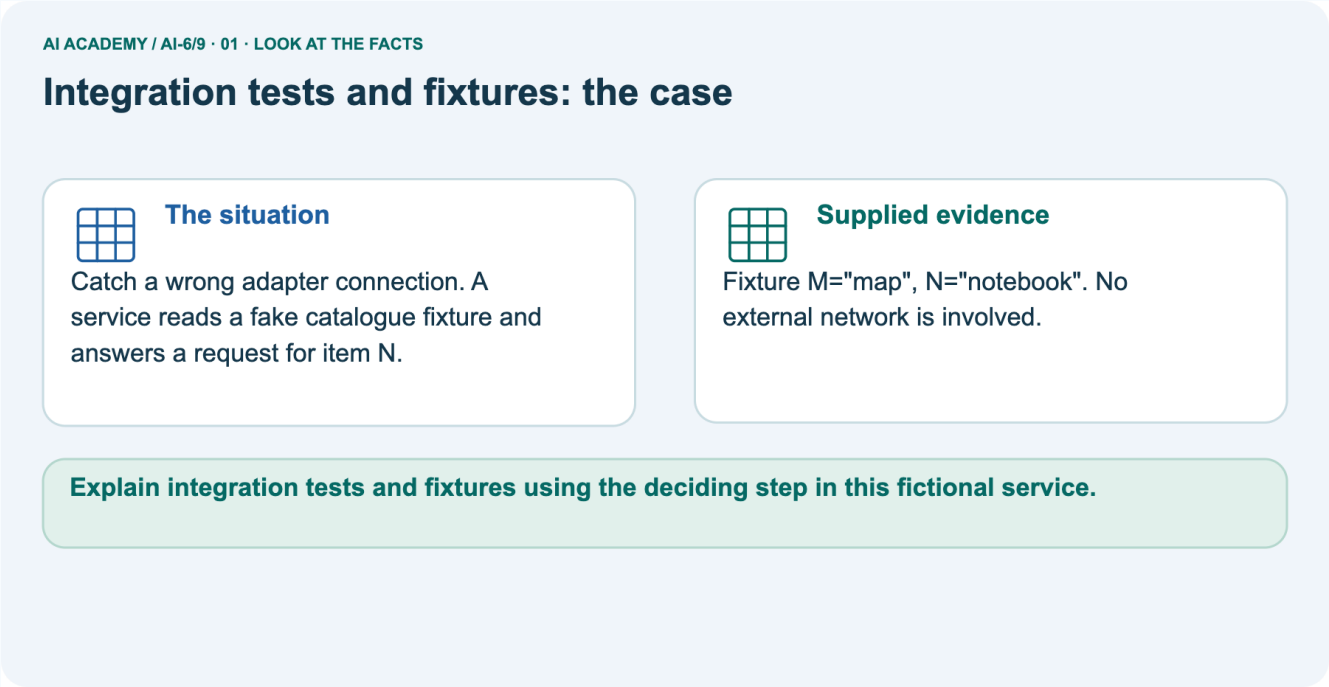
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Integration tests and fixtures: the case



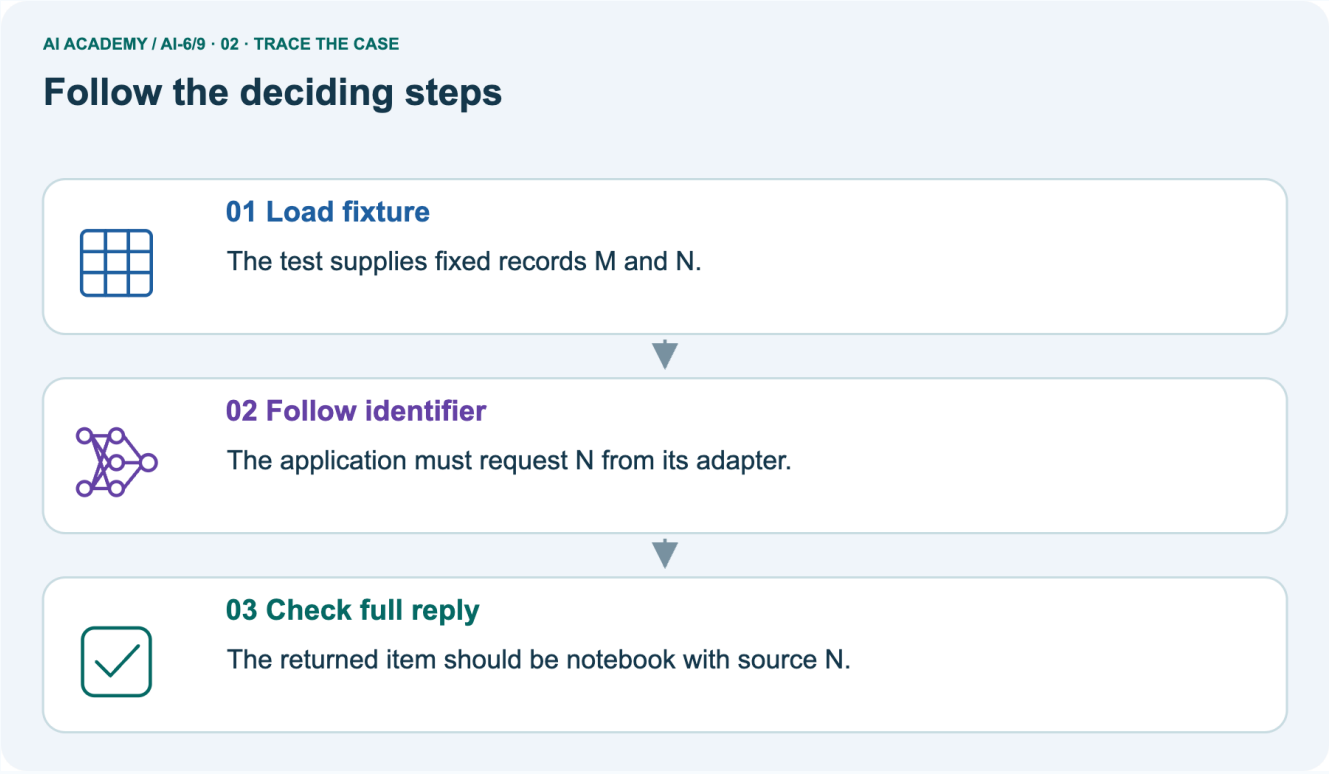
Follow the labelled picture with the matching explanation. Open the vector diagram

Catch a wrong adapter connection. A service reads a fake catalogue **fixture** and answers a request for item N. Supplied facts: **Fixture** M="map", N="notebook". No external network is involved.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



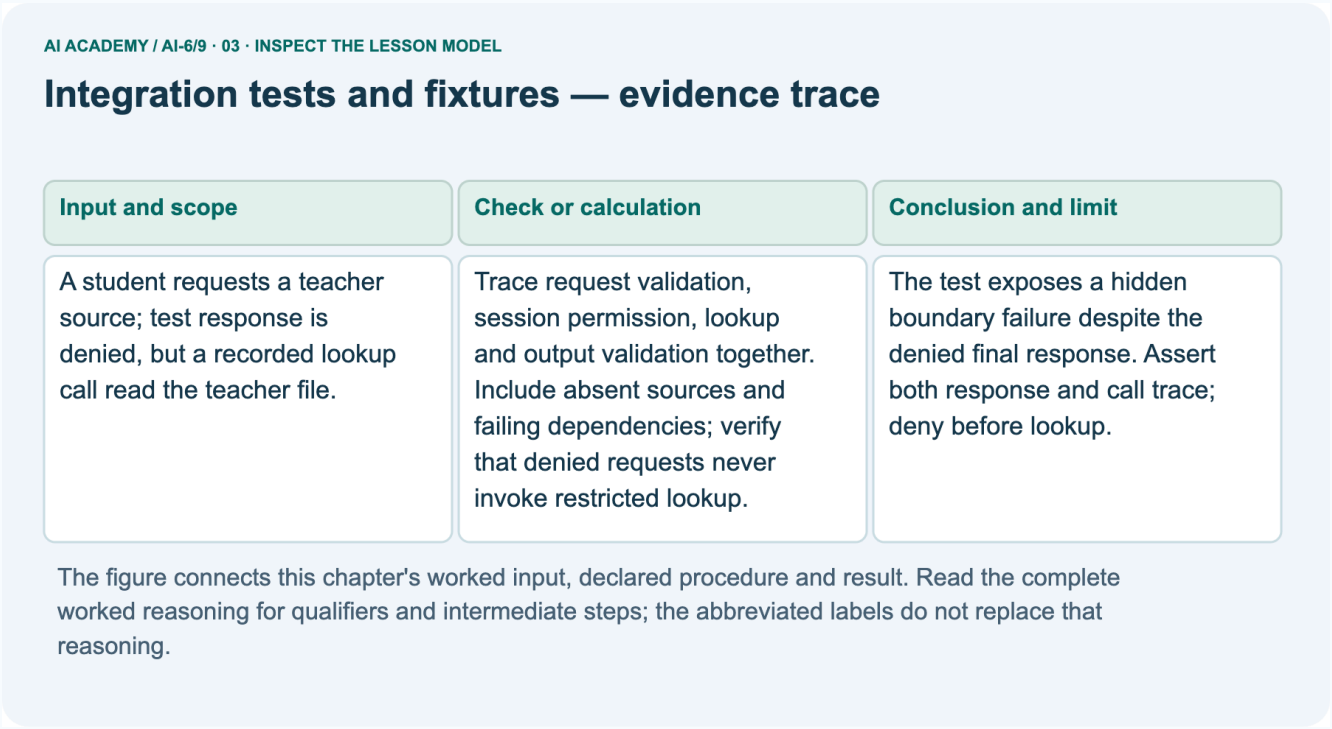
Follow the labelled picture with the matching explanation. Open the vector diagram

Load **fixture**: The test supplies fixed records M and N. → Follow identifier: The application must request N from its adapter. → Check full reply: The returned item should be notebook with source N.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Integration tests and fixtures — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Integration tests and **fixture**s — **evidence** trace.

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. | Trace request validation, session permission, lookup and **output** validation together.

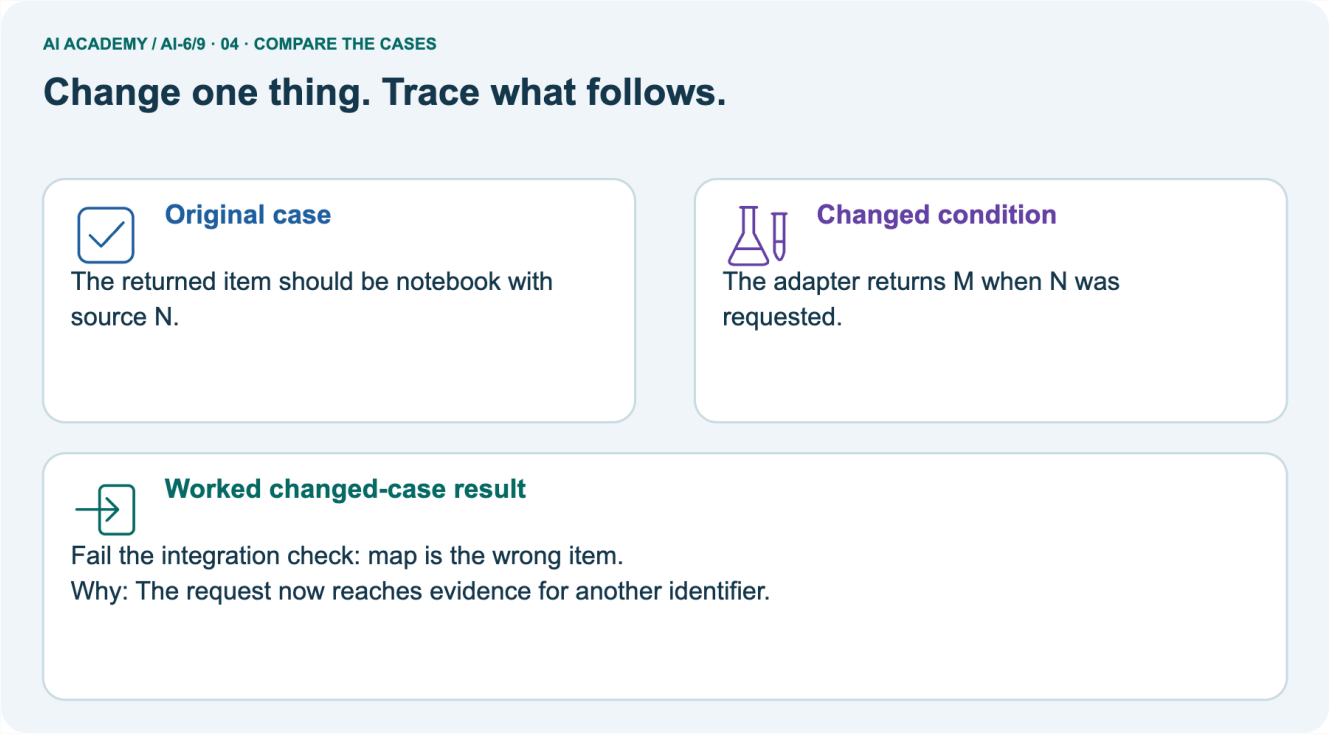
Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup. | The test exposes a hidden boundary failure despite the denied final response.

Assert both response and **call trace**; deny before lookup.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

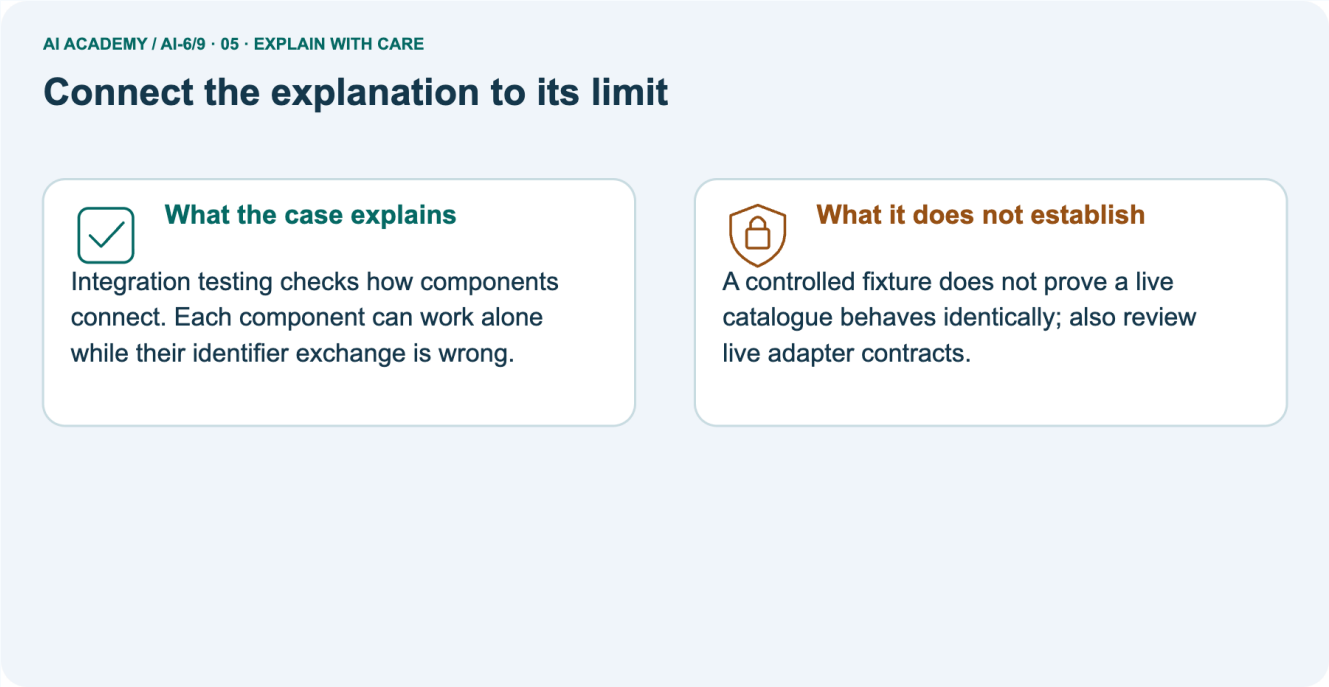
Original case: The returned item should be notebook with source N. Changed condition: The adapter returns M when N was requested. Expected result: Fail the integration check: map is the wrong item.

Reason: The request now reaches **evidence** for another identifier.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

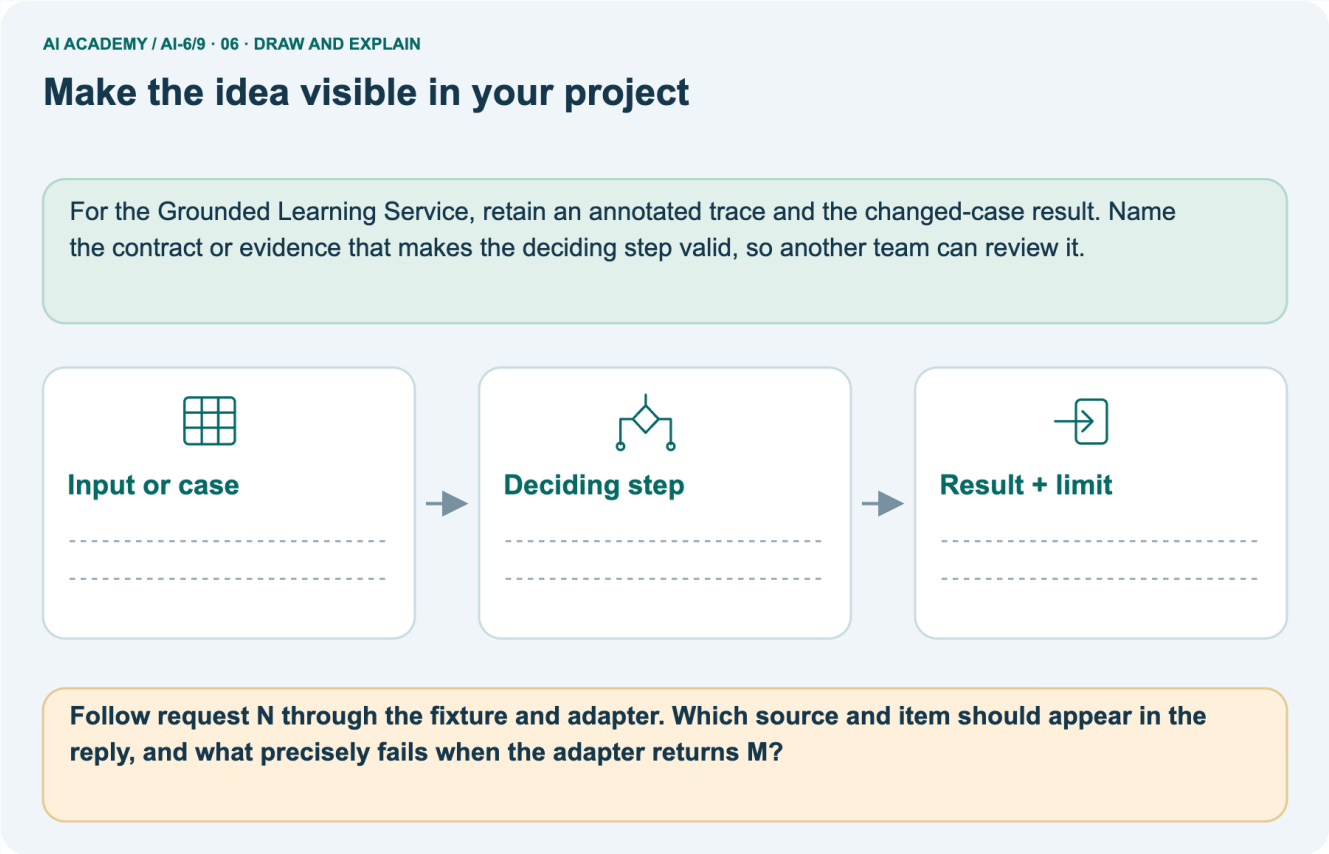
Integration testing checks how components connect. Each component can work alone while their identifier exchange is wrong.

Limit: A controlled **fixture** does not prove a live catalogue behaves identically; also review live adapter contracts.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Follow request N through the **fixture** and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Integration tests and fixtures: the case

02. Follow the deciding steps

03. Integration tests and fixtures — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.